

HDOC Day-28

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Batch	15

Question 1

Write a C program to: (a) read positive rows and columns, repeating input until both are positive; (b) declare a variable-length 2D integer array; (c) read elements column by column using nested loops; and (d) print the elements in row-major order using a single loop and a linear index.

Algorithm

1. Read rows and columns until both are positive.
2. Declare a variable-length array `a[rows][cols]`.
3. Use the outer loop for columns and inner loop for rows to input elements.
4. Use one loop from 0 to `rows*cols-1`.
5. Convert the linear index `k` to row `k/cols` and column `k%cols` and print the element.
6. Stop.

C Program

```
#include <stdio.h>

int main() {
    int rows, cols, i, j, k;

    do {
        printf("Enter rows and columns: ");
        scanf("%d %d", &rows, &cols);
        if (rows <= 0 || cols <= 0)
            printf("Please enter positive values.\n");
    } while (rows <= 0 || cols <= 0);

    int a[rows][cols];

    printf("Enter array elements column by column:\n");
    for (j = 0; j < cols; j++) {
        for (i = 0; i < rows; i++) {
            printf("a[%d][%d] = ", i, j);
            scanf("%d", &a[i][j]);
        }
    }

    printf("\nElements in row-major order:\n");
    for (k = 0; k < rows * cols; k++)
        printf("%d ", a[k / cols][k % cols]);

    printf("\n");
    return 0;
}
```

Sample Output

```
Enter rows and columns: 2 3
Enter array elements column by column:
a[0][0] = 1
a[1][0] = 4
a[0][1] = 2
a[1][1] = 5
a[0][2] = 3
a[1][2] = 6

Elements in row-major order:
1 2 3 4 5 6
```

Test Cases

Input rows x cols	Input elements	Expected row-major output
2 x 3	1,4,2,5,3,6	1 2 3 4 5 6
-2 x 3	-	Ask again for positive values

Question 2

Write a C program that reads elements into a variable-length 2D array, copies them into a 1D array in row-major order, and displays both arrays.

Algorithm

1. Read positive rows and columns.
2. Declare VLA `a[rows][cols]` and one-dimensional array `b[rows*cols]`.
3. Read all elements of the 2D array.
4. Copy each element into `b` using `b[i*cols+j] = a[i][j]`.
5. Display the 2D array and then the 1D array.
6. Stop.

C Program

```
#include <stdio.h>

int main() {
    int rows, cols, i, j;

    do {
        printf("Enter rows and columns: ");
        scanf("%d %d", &rows, &cols);
    } while (rows <= 0 || cols <= 0);

    int a[rows][cols];
    int b[rows * cols];

    printf("Enter elements of 2D array:\n");
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            scanf("%d", &a[i][j]);
            b[i * cols + j] = a[i][j];
        }
    }

    printf("\n2D Array:\n");
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++)
            printf("%d ", a[i][j]);
        printf("\n");
    }

    printf("\n1D Array:\n");
    for (i = 0; i < rows * cols; i++)
        printf("%d ", b[i]);

    printf("\n");
    return 0;
}
```

Sample Output

```
Enter rows and columns: 2 3
Enter elements of 2D array:
1 2 3
4 5 6

2D Array:
1 2 3
4 5 6

1D Array:
1 2 3 4 5 6
```

Test Cases

Rows x cols	2D input	Expected 1D array
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2 x 3	1 2 3 / 4 5 6	1 2 3 4 5 6
2 x 2	7 8 / 9 10	7 8 9 10